PIPINGPRO ACADEMY

Free Guide

Introduction to Piping

Pipe Materials · Fittings · Pipe Support · Stress Basics

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30+ years · 6 countries · Oil & Gas

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Introduction to Piping

In the long history of human civilization, the design of piping networks has been practiced for thousands of years. The earliest piping systems were used for irrigation in agriculture; for example, people in China used bamboo pipes as early as 3000–2000 BC. Over time, the use of pipes expanded into many aspects of daily life. Rapid technological development in piping only occurred in the 19th century.

A major advance came in 1934 when R. H. Tingey published his article Method of Calculation Thermal Expansion Stresses in Piping, introducing the concept of the virtual center of gravity or elastic center. He developed early methods for calculating thermal expansion stresses and reactions in piping systems, particularly focusing on three-dimensional pipe bends, elbows, and flexibility analysis.

Another milestone occurred in 1932 when the engineering department of Standard Oil Company (Indiana) issued an internal report explaining how flexibility factors could be derived from basic circular shapes. This work was later included in the first edition of Design of Piping Systems by The M. W. Kellogg Company (1941).

Key Point

In general, materials used for pipes are divided into two main categories: Metallic (Ferrous and Non-Ferrous) and Non-Metallic (Plastic and Non-Plastic).

1. Pipe Materials

Understanding pipe material selection is one of the most fundamental skills for a piping engineer. The wrong material choice can lead to corrosion failures, safety incidents, and costly replacements. This section covers the full range of materials you will encounter in Oil & Gas projects.

1.1 Ferrous Pipe Materials

Cast Iron Pipe

Cast Iron is an alloy of iron and carbon with a carbon weight content of more than 2% by weight. It is often combined with other elements to increase corrosion resistance, such as Silicon, Nickel, or Chromium. It is mainly used for water distribution and sewage applications.

Ductile Iron

Ductile Iron is stronger and tougher than Cast Iron. It is a high-carbon magnesium-treated product containing graphite in the form of spheroids. Like Cast Iron, Ductile Iron is used for sewage service but can also be used for potable water.

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Carbon Steel Pipe

This is the most widely used material in the Oil and Gas industry due to wide availability, high strength, and a huge range of compatible fittings and connection systems. Carbon Steel pipe has been used for oil and gas services both above and underground.

Why Carbon Steel Dominates

Wide availability globally · High strength-to-cost ratio · Extensive fitting and valve compatibility · Proven track record in Oil & Gas for over a century.

Stainless Steel Pipe

Stainless Steel pipes are used whenever corrosion resistance is needed. Adding Chromium provides good corrosion resistance. Three main types exist:

- Austenitic Stainless Steel (most common — grades 304 and 316)
- Ferritic Stainless Steel
- Martensitic Stainless Steel

Austenitic grades with lower carbon content are designated with the suffix 'L' (e.g. 316L). Stainless steel is rarely used for underground applications due to cost.

1.2 Non-Ferrous Pipe Materials

Aluminum Alloy Pipe

Aluminum alloys have a density of approximately one-third that of steel and are malleable, weldable, and corrosion-resistant. They are not common in piping systems but are sometimes found in compressed gas applications. Relevant standards include ASTM B210 and ASTM B241.

Copper Tubing

Copper, Bronze, and Brass have long been used in pipes due to their corrosion-resistant properties, primarily for water applications. Copper Tubing is the most widely used material for plumbing, hydronics, and refrigeration equipment.

1.3 Non-Metallic Pipe Materials

Plastic Pipe — Thermoplastics and Thermosetting

Plastic pipes consist of long, heavy organic compound molecules. Two main forms exist:

- **Thermoplastics:** Thermoplastics

  Can be repeatedly softened when heated and hardened when cooled without affecting material properties.
- **Thermosetting:** Thermosetting materials

  Irreversibly cured into a permanent shape during manufacture. Cannot be reshaped by heating.

Plastic pipes are particularly well-suited for inorganic fluids such as diluted acids, ammonia gases, alkalis, salts, organic fluids, natural gas, and oils.

Non-Plastic Pipes — Concrete, Clay, Glass

Non-plastic pipe includes reinforced concrete pipe, clay pipe, wood pipe, and glass pipe:

- Clay pipes — made from shale and clay; used for stormwater, sewage, and industrial waste. Robust and eco-friendly — some in use for over 200 years.
- Concrete pipe — used in pressure and gravity applications, storm and sanitary sewers.
- Glass pipes — used to withstand chemical reactions; common in chemical and pharmaceutical industries.

1.4 Corrosion Resistant Alloy (CRA) Pipe

CRA pipes have become increasingly important as a replacement for carbon steel in applications where high corrosivity is expected — particularly for sour gas and oil, saltwater pipelines, and produced fluids with high water cuts and concentrations of CO_2 and H_2S.

Two major types of CRA:

1. Solid CRA line pipe
2. CRA Weld Overlay (clad or lined)

Industry Trend

The demand for CRA clad and lined pipes is increasing due to anticipated rises in corrosive conditions — driven by higher water cuts and CO_2/H_2S concentrations in produced fluids transported over longer distances at higher pressures.

2. Pipes, Fittings & Flanges

2.1 Pipe Size and Wall Thickness

A pipe is a tube with a round cross-section defined by its Outside Diameter (OD), wall thickness (t), and resulting Inside Diameter (ID). Pipe dimensions must conform to:

3. ASME B36.10-2022: Welded and Seamless Wrought Steel Pipe
4. ASME B36.19-2022: Welded and Seamless Wrought Stainless-Steel Pipe

Two common pipe designation systems are used:

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- NPS (Nominal Pipe Size): A dimensionless designator. For NPS 14 and larger, the OD equals the NPS value. For smaller sizes (NPS 12 and below), the OD is larger than the nominal size.
- DN (Diamètre Nominal): The metric equivalent, representing approximate inside diameter in millimetres (e.g. DN 50 ≈ 50 mm ID).

Pipes are generally grouped as:

- Large Bore Pipe: Pipes larger than 2 inches
- Small Bore Pipe: Pipes 2 inches and smaller
- Tubing: Up to 4 inches with thinner walls than standard pipe

Practical Note

Pipe sizes 3½" and 5" are rarely used in industrial projects because of limited fitting and valve availability. Always check fitting compatibility when selecting pipe size.

Wall Thickness — Schedule Numbers

Pipe wall thickness is specified by a Schedule Number rather than actual thickness. Originally three designations existed: Standard (STD), Extra Strong (XS), and Double Extra Strong (XXS). Schedule numbers now run from 5S, 10, 10S, 20, 30, 40, 60, 80, 100, 120, 140, to 160.

2.2 Pipe Formulae

Piping engineers frequently need pipe properties for stress analysis, support design, and foundation calculations:

- Inside Diameter: $ID = OD - 2t$
- Area of Metal (in^2): $Am = 0.7854\ (OD^2 - ID^2)$
- Moment of Inertia (in^4): $I = 0.0491\ (OD^4 - ID^4)$
- Weight of Pipe per foot (lb): $= 10.6802 \times t \times (D - t)$
- Weight of Water per foot (lb): $= 0.3405 \times ID^2$

2.3 Pipe Fittings

The piping system requires components beyond straight pipe — these are called fittings, made from the same material as the connecting pipe.

Elbow

Used where the pipe changes direction of travel (e.g. from west to south). Available in 45° and 90° long radius (LR) and short radius (SR) types.

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Reducer

Changes pipe diameter. Two types: Concentric Reducer (centrelines aligned) and Eccentric Reducer (flat-top or flat-bottom, depending on installation location).

Tee

Divides or combines flow. Equal Tee (same diameter throughout) or Reducing Tee (different inlet/outlet diameters).

Olet

Used for branching at 90° or 45° from a straight pipe. A popular alternative to Tee for smaller branch connections. Common types include Weldolet, Sockolet, and Threadolet.

Cap

Covers the end of a pipe — same function as a plug, installed at the termination of a piping run.

Stub-In

A direct branch connection welded into the run pipe without a fitting. Cost-effective but limited by temperature, pressure, and fluid type. Can be installed with or without a Reinforcing Pad.

2.4 Pipe Flanges

A flange is a pipe connection used to join pipes, equipment nozzles, or other components — particularly where regular maintenance or disassembly is required. Unlike fittings, flanges are generally made through the forging process and have a machined face surface.

Flange Rating (Class)

Flanges are grouped into Classes based on Design Pressure and operating temperature:

Class 150 | 300 | 400 | 600 | 900 | 1500 | 2500

The Flange Rating is expressed in psi but commonly referred to as 'pound' — a 300-pound flange does NOT mean the design pressure is 300 psi. The actual allowable pressure depends on the material group and operating temperature per ASME B16.5.

3. Pipe Support

Pipe support is critical for the safe operation of piping systems. Due to the slender nature of piping, it often cannot be self-supporting and may require restraint or bracing against thermal expansion, vibration, and other effects. Selecting optimal support locations and economical types is one of the most challenging aspects of piping design.

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Key Design Principle

Supports shall be located as close as possible to structural steel — avoid proximity to valves, welds, or other obstructions — with adequate space for the chosen component type.

Three types of loads normally occur at the point of support:

5. Vertical Loads — from pipe weight, insulation, and fluid
6. Lateral Loads — forces acting from the side of the pipe
7. Axial Loads — forces in the direction of the pipe (longitudinal)

3.1 Allowable Pipe Span

The Allowable Pipe Span is the maximum distance between two consecutive pipe supports. Correct span selection is critical for:

8. Preventing excessive stress (keeping stress within the allowable limit of the pipe material)
9. Preventing excessive deflection — avoiding pockets and interference with adjacent pipes
10. Controlling natural frequency to avoid vibration resonance

3.2 Standard Pipe Support

Standard pipe support types are designed to be used repeatedly across different locations without changing their shape or design. They are pre-engineered to withstand loads up to certain limits and are referenced in MSS SP-58.

3.3 Special Pipe Support

Special pipe supports are specifically designed when standard support cannot be used due to:

11. The pipe size to be supported exceeds the maximum size of available standard support
12. The applied load exceeds the maximum capacity of the standard type
13. The pipe location is unusual — far from existing structure — making standard support impractical

Special supports may use standard supports as a design basis and modify them for the specific requirement. Design typically references MSS SP-58. For complex configurations, the work is often handed over to the Civil Structural Engineering group.

4. Pipe Stress: The Basics

The main purpose of performing Piping Stress Analysis is to ensure that the piping system has sufficient flexibility to:

14. Accommodate thermal expansion or contraction
15. Prevent excessive movement at pipe support locations
16. Prevent excessive movement at nozzle connections or anchor points

Insufficient flexibility can cause:

- Failure from overstress or fatigue of the piping material
- Excessive stresses in pipe supports
- Leakage at joints
- Detrimental distortion or overload of connected equipment nozzles
- Resonance with imposed vibrations

4.1 Methods of Analysis

ASME B31.3-2024 paragraph 319.4 defines two methods:

17. No Formal Analysis Required — for simple, flexible, low-temperature systems meeting the code criteria
18. Formal Analysis Required — using computer software such as CAESAR II for critical systems

4.2 Critical Line List

The Critical Line List identifies which piping systems require formal computer analysis and which systems can be assessed informally — ensuring all piping meets the applicable Codes and Standards requirements.

4.3 Pipe Loading Types

Sustained Loads

Sustained loads include dead loads (pipes, flanges, fittings, valves) and live loads (internal fluid, snow, ice) as well as pressure, wind, and earthquake. The key characteristic of sustained loads: they can cause gross collapse of the piping system.

Thermal Loads

Thermal loads arise from forces and moments caused by resistance to free thermal expansion or contraction due to restraints and anchors. Key characteristic: thermal loads are cyclic and disappear when system restraints are removed.

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4.4 Stress Categories

Primary Stress

Primary Stress is caused by Sustained Load. It is not self-limiting — if it exceeds the yield strength of the material through the entire pipe cross-section, it will cause failure. This is the most hazardous stress category in piping analysis.

Secondary Stress

Secondary Stress is caused by thermal loads — the expansion or contraction of the pipe due to operating temperature. It must satisfy an imposed strain pattern rather than equilibrium with an external load. Key characteristic: secondary stress is self-limiting.

Summary

Primary stress is caused by weight and pressure — it can cause catastrophic failure. Secondary stress is caused by thermal expansion — it is self-limiting but can cause fatigue over many cycles. Both must be kept within the allowable limits defined by the applicable code (ASME B31.3 for process piping).

What's Next?

This free guide has introduced you to the foundation of piping engineering — pipe materials, fittings and flanges, pipe support, and the basics of stress analysis. These concepts underpin everything you will do as a piping or pipeline engineer.

The full PipingPro Academy course covers four comprehensive modules:

19. Pipes, Fittings, Pipe Support & Pipe Material (this guide — expanded with worked examples)
20. Fundamentals of Piping Stress Analysis — load cases, SIF, ASME B31.3
21. Onshore Pipeline Design — wall thickness, hoop stress, ASME B31.4 & B31.8
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